

Treatment Success and Failure Prediction using Machine Learning Models for Pulmonary Tuberculosis Patients under the DOH-CAR TB-DOTS Program

Author Names

Aaron Kyle M. Aguilar¹, Mark Lestat C. Agustin², Keanu Sonn M. Fortaleza³, Benny Gil A. Lactaotao⁴,
Marven Joffre G. Luis⁵, Georcelle Leisley U. Nuarin⁶, Sean Drei A. Octavo⁷, Gebreyil Isaac P. Rabang⁸

Affiliations

^{1 2 3 4 5 6 7 8} Department of Computing and Information Studies, Saint Louis University, Baguio City, Philippines

Corresponding Author:

Josephine S. Dela Cruz, DIT – jdelacruz@slu.edu.ph

Abstract

This study proposes a multi-layer analytical framework using machine learning to predict pulmonary tuberculosis (PTB) treatment success or failure in the Cordillera Administrative Region (CAR), Philippines. Utilizing demographic, clinical, and diagnostic data from 2015 to 2025, supervised models (Logistic Regression, Support Vector Machines, and Extreme Gradient Boosting) and temporal models (Recurrent Neural Networks and Long Short-Term Memory) are trained to identify risk factors. Shapley Additive Explanations (SHAP) ensure interpretability. The models will be integrated into a clinical decision support tool for early intervention. Aligning with the Sustainable Development Goals for good health and well-being, this predictive tool aims to enhance TB control efforts, optimize resource allocation, and improve patient management.

Keywords

Clinical decision support, machine learning, predictive modeling, pulmonary tuberculosis, treatment outcomes

1. Introduction

Pulmonary Tuberculosis (PTB) is a chronic infectious disease that continues to challenge public health due to its gradual progression and risk of treatment failure (Tobin & Tristram, 2024). In the Philippines, TB remains a major public health concern, with the country being one of the top contributors to the global TB burden (World Health Organization, 2022). In the Cordillera Administrative Region (CAR), the Department of Health (DOH) reported 437 confirmed TB cases in Baguio City alone in 2022, alongside an estimated 4,405 missing or unreported cases (Department of Health Republic of the Philippines, n.d.).

While multiple studies have examined risk factors such as sociodemographic characteristics, comorbidities, and medical history, findings remain fragmented as traditional statistical analyses often assume linear relationships and examine only a few factors simultaneously (Amante &

Abdosh, 2015). To address this gap, this study adopts an integrative machine learning framework to combine multiple dimensions of data in analyzing interrelated variables.

The specific research objectives of this study are as follows:

1. **Data Pre-processing & Feature Engineering:** To clean, pre-process, and digitalize PTB patient records, creating engineered features that ensure the dataset is optimized for statistical analysis and machine learning applications.
2. **Statistical Analysis & Profiling:** To perform descriptive and inferential statistics to profile the demographic, clinical, and treatment characteristics of the study population, identifying baseline variables that significantly influence treatment outcomes.
3. **AI Modeling & Comparative Analysis:** To develop, validate, and evaluate both non-temporal and temporal machine learning models for predicting PTB treatment outcomes, comparing the predictive performance of static baseline data against dynamic, time-dependent patient patterns throughout the course of therapy.
4. **Data Visualization & Clinical Interpretation:** To visualize patient risk levels, statistical trends, and predictive outputs from the AI models, providing clear insights to support clinical interpretation and the early identification of individuals at risk of poor outcomes.

2. Methodology

This study utilizes a retrospective dataset of PTB patients from the CAR covering the period from 2015 to 2025, consolidated by the DOH-CAR. Data undergoes a standardized preprocessing pipeline, including schema harmonization, error correction, and missing value imputation via Multivariate Imputation by Chained Equations, or MICE (Alruhaymi & Kim, 2021).

To predict treatment outcomes defined by the National Tuberculosis Program criteria, supervised models including Logistic Regression, Support Vector Machines (SVM), and Extreme Gradient Boosting (XGBoost) are utilized. Furthermore, temporal models using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) are employed to capture longitudinal treatment dynamics (Lu et al., 2022; Nayebi et al., 2022). Model performance is assessed using standard classification metrics, and Shapley Additive Explanations (SHAP) are applied to ensure clinical interpretability (Peng et al., 2024).

3. Results and Discussion

This section presents the results of machine learning analyses aimed at predicting treatment success and failure among pulmonary tuberculosis patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes, while predictive model performance is evaluated using test and validation datasets. Both static (Random Forest, Gradient Boosting, LightGBM, XGBoost, Logistic Regression) and temporal models are compared, and feature importance analyses provide insights into the key factors influencing treatment outcomes. The discussion interprets these findings in

the context of clinical relevance and public health priorities, highlighting implications for early intervention, resource allocation, and strengthened TB management.

3.1 Dataset Characteristics and Class Distribution

The final dataset exhibited significant class imbalance, with 86.2% of cases classified as treatment success and 13.8% as treatment failure. To ensure robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors, including four newly introduced geographic and facility-level variables and one derived clinical delay variable (Diagnosis_to_Treatment_days). Resampling using SMOTE combined with Edited Nearest Neighbors (SMOTE+ENN) was applied exclusively to the training set, increasing the minority (failure) class by 433% and achieving a near-balanced 0.70:1 ratio. This approach mitigated class bias while preserving realistic evaluation conditions in the test and validation sets.

3.2 Validation Set Performance and Generalization

Table 1: Test Set Model Performance

Model	AUC (Area Under the Curve)	Recall (Failure)	F1 (Failure)
Random Forest	0.7154	57.98%	0.3539
Gradient Boosting	0.7121	44.68%	0.3262
LightGBM	0.7119	48.40%	0.3467
XGBoost	0.7095	47.87%	0.3416

Among all evaluated models, Random Forest achieved the highest overall performance on the test set, with an AUC of 0.7154 and an F1-score of 0.3539 for the failure class. It also demonstrated balanced recall (57.98%) without excessive false positives. Gradient Boosting, LightGBM, and XGBoost achieved comparable AUC values (~0.71), though with slightly lower recall for treatment failure. Logistic Regression demonstrated the highest recall (77.13%); however, its precision was notably low (19.6%), resulting in over-alerting and reduced overall accuracy (~51%). These findings suggest that while Logistic Regression aggressively identifies high-risk patients, it lacks specificity, potentially burdening healthcare systems with excessive false alarms.

On the 10% held-out validation set, Random Forest maintained superior performance, achieving a validation AUC of 0.7380 and recall of 52.13% for treatment failure. Notably, all models demonstrated positive AUC gains from test to validation sets, indicating strong generalization and absence of overfitting. The consistency of Random Forest across both evaluation phases confirms its robustness and suitability for deployment within a clinical decision support framework.

3.3 Validation Set Performance and Generalization

Table 2: Test Set Performance Metrics of Temporal Models for TB Treatment Outcome Prediction

Metric	Bi-LSTM Baseline	Bi-LSTM Augmented	XGBoost Baseline	XGBoost SMOTE-ENN	LightGBM SMOTE-ENN	Random Forest SMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
Specificity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419
Threshold	0.15	0.86	0.10	0.34	0.43	0.70

Table 1 presents the performance metrics of all temporal models evaluated on the held-out test set. Both Bi-LSTM variants and XGBoost Baseline achieved the highest overall accuracy (0.9048) and F1 score (0.9500), reflecting their ability to correctly predict the dominant Success class. Precision values were similarly high (0.9048), indicating few false positives for the majority class. However, recall for the minority Failure class reached 1.0 for these models, while specificity remained 0.0, demonstrating that no actual Failure cases were correctly identified in the small test set. Tree-based models augmented with SMOTE-ENN, particularly Random Forest, showed a more balanced performance. Random Forest achieved slightly lower overall accuracy (0.7143) but higher recall for the Failure class (0.7895) and competitive PR-AUC (0.9419), indicating better identification of high-risk patients. ROC-AUC scores further illustrate differences in probability calibration, with XGBoost Baseline achieving the highest ROC-AUC (0.5789), while the Bi-LSTM Augmented variant performed lower (0.3684), likely due to overfitting on limited Failure samples. Overall, these

results suggest that while Bi-LSTM and XGBoost excel at predicting the majority outcome, Random Forest with SMOTE-ENN offers the most effective detection of treatment failure, making it more suitable for clinical decision support applications.

4. Conclusions and Recommendations

This study investigated the use of machine learning models to predict treatment outcomes—Success or Failure—of pulmonary tuberculosis patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. Both temporal models, specifically the Bi-LSTM Baseline and Augmented variants, and tree-based models, including XGBoost, LightGBM, and Random Forest with SMOTE-ENN, were evaluated on a stratified 70/20/10 patient-level split, incorporating progressive temporal samples from treatment months M0 through M12. The test set performance revealed that while Bi-LSTM and XGBoost Baseline models achieved high overall accuracy (0.9048) and F1 scores (0.9500), their ability to detect treatment failures was limited, as indicated by zero specificity for the minority Failure class. In contrast, Random Forest with SMOTE-ENN demonstrated more balanced performance, achieving a recall of 0.7895 for treatment failure and a high PR-AUC (0.9419), indicating greater reliability in identifying patients at risk of adverse outcomes. These results highlight that tree-based models augmented with SMOTE-ENN are particularly effective at handling severe class imbalance, while temporal models provide strong predictive accuracy for the majority outcome.

The study contributes in several ways. By integrating temporal monitoring data with static baseline features, it demonstrates that patient-level longitudinal information improves predictive performance over static data alone. The application of SMOTE-ENN and advanced data augmentation techniques for Bi-LSTM effectively addresses the extreme class imbalance inherent in TB treatment outcomes, enhancing the model's capacity to detect high-risk patients. Furthermore, the findings suggest that predictive models, particularly Random Forest with SMOTE-ENN, have practical value for clinical decision support systems, enabling clinicians to identify patients at risk of treatment failure early and prioritize interventions accordingly.

Based on these findings, it is recommended that healthcare facilities and policymakers consider integrating machine learning models like Random Forest with SMOTE-ENN into existing TB monitoring programs to improve treatment outcomes and optimize resource allocation. Future research should focus on larger, multi-region datasets to enhance generalization, explore hybrid modeling approaches combining deep learning and tree-based methods, and investigate additional features such as social determinants of health and geographic risk factors. Additionally, enhancing model interpretability, particularly of Bi-LSTM attention outputs, could improve clinician trust and adoption.

Overall, this study aligns with Sustainable Development Goal 3 – Good Health and Well-Being by providing a data-driven approach to improving TB treatment outcomes. By accurately identifying patients at risk of treatment failure, these predictive models have the potential to reduce TB-related morbidity and mortality, inform targeted public health interventions, and contribute to more efficient and effective TB care within the Cordillera Administrative Region.

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